

Homogeneous Transformation in Planar and Spatial Mechanisms using Python Robotics Toolbox

Course Name

Introduction to Robotics Laboratory

Duration

2 Hours

Software Required

- Python 3.10 or later
 - Robotics Toolbox for Python
 - SpatialMath Python Package
 - NumPy
 - Matplotlib
 - Jupyter Notebook / VS Code
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Learning Objectives

At the end of this laboratory session, students will be able to:

1. Understand the concept of homogeneous transformation matrices.
 2. Perform translation in planar (2D) and spatial (3D) coordinate systems.
 3. Perform rotation in planar and spatial coordinate systems.
 4. Combine translation and rotation into a single homogeneous transformation.
 5. Compare different orders of transformation.
 6. Visualize transformed coordinate frames using Python Robotics Toolbox.
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Prerequisite Knowledge

Students should be familiar with:

- Cartesian Coordinate System
 - Matrix Multiplication
 - Rotation Matrix
 - Basic Python Programming
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Theory

A robot manipulator moves in space by changing its position and orientation.

Instead of handling rotation and translation separately, robotics combines both into a **Homogeneous Transformation Matrix (HTM)**.

The homogeneous transformation matrix is represented as

$$T = \begin{bmatrix} R & P \\ 0 & 1 \end{bmatrix}$$

where

- **R** = Rotation Matrix
- **P** = Position Vector

The Robotics Toolbox provides

- **SE2** → Planar Transformations (2D)
- **SE3** → Spatial Transformations (3D)

Laboratory Exercise

Part A – Planar Mechanism (2D)

Task 1 – Translation

Objective

Create a translation in the XY plane.

Python Code

```
from spatialmath import SE2
```

```
T = SE2(2,3)
```

```
print(T)
```

Student Activity

Run the program for the following translations.

Trial X Y

```
1    2    3
```

Trial X Y

2 5 4
3 3 -2
4 -4 6

Observation Table**Trial Translation Matrix Obtained**

1
2
3
4

Task 2 – Rotation**Objective**

Create a rotation matrix.

Python Code

```
from spatialmath import SE2
```

```
R = SE2.Rot(45,"deg")
```

```
print(R)
```

Student Activity

Repeat the program for

- 30°
 - 60°
 - 90°
 - 180°
-

Observation Table

Rotation Transformation Matrix

30°

60°

90°

180°

Task 3 – Translation followed by Rotation

Objective

Combine translation and rotation.

Python Code

```
from spatialmath import SE2
```

```
T = SE2(4,2)
```

```
R = SE2.Rot(90,"deg")
```

```
H = T * R
```

```
print(H)
```

Student Activity

Complete the following table.

Translation	Rotation
(3,1)	45°
(5,2)	60°
(2,6)	120°

Record the Final Matrix

Case Transformation Matrix

1

2

3

Task 4 – Rotation followed by Translation

Objective

Compare the order of transformations.

Python Code

```
from spatialmath import SE2
```

```
T = SE2(4,2)
```

```
R = SE2.Rot(90,"deg")
```

```
print(T*R)
```

```
print(R*T)
```

Questions

1. Are the two matrices identical?
 2. Which coordinate changed?
 3. Why does the order of multiplication matter?
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Part B – Spatial Mechanism (3D)

Task 5 – Translation in 3D

Objective

Create a translation in XYZ space.

Python Code

```
from spatialmath import SE3
```

```
T = SE3(2,3,4)
```

```
print(T)
```

Student Activity

Repeat for

X Y Z

4 2 1

5 5 5

-2 3 7

Observation

Record all transformation matrices.

Task 6 – Rotation about X-axis

Objective

Perform rotation about X-axis.

Python Code

```
from spatialmath import SE3
```

```
Rx = SE3.Rx(45,"deg")
```

```
print(Rx)
```

Student Activity

Repeat for

- 30°
- 90°

- 180°

Observation

Record the matrices.

Task 7 – Rotation about Y-axis**Objective**

Rotate about Y-axis.

Python Code

```
from spatialmath import SE3
```

```
Ry = SE3.Ry(60,"deg")
```

```
print(Ry)
```

Student Activity

Perform

- 45°
 - 120°
 - 180°
-

Task 8 – Rotation about Z-axis**Objective**

Rotate about Z-axis.

Python Code

```
from spatialmath import SE3
```

```
Rz = SE3.Rz(30,"deg")
```

```
print(Rz)
```

Student Activity

Repeat for

- 60°
 - 90°
 - 270°
-

Task 9 – Combined Translation and Rotation

Objective

Perform multiple transformations.

Python Code

```
from spatialmath import SE3
```

```
T = SE3(2,3,4)
```

```
Rx = SE3.Rx(30,"deg")
```

```
Ry = SE3.Ry(45,"deg")
```

```
Rz = SE3.Rz(60,"deg")
```

```
H = T * Rx * Ry * Rz
```

```
print(H)
```

Student Activity

Repeat using

Translation Rotations

(5,4,2) $30^\circ, 45^\circ, 60^\circ$

(2,8,3) $45^\circ, 30^\circ, 90^\circ$

(1,5,4) $60^\circ, 60^\circ, 60^\circ$

Observation

Write the obtained transformation matrices.

Task 10 – Coordinate Frame Visualization

Objective

Visualize transformed coordinate frames.

Python Code

```
from spatialmath import SE3
```

```
T = SE3(2,3,4) * SE3.Rz(45,"deg")
```

```
T.plot(frame='A')
```

Student Activity

Plot coordinate frames for

- Frame A
- Frame B
- Frame C

Capture screenshots and paste them in the report.

Challenge Exercise (20 Minutes)

A robot end-effector is initially located at the origin.

Perform the following operations sequentially:

1. Translate by (3,2,1)
 2. Rotate 45° about X-axis
 3. Rotate 60° about Y-axis
 4. Rotate 30° about Z-axis
 5. Translate by (1,2,0)
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Write a Python Program to

- Generate the final homogeneous transformation matrix.
 - Determine the final position of the end-effector.
 - Plot the transformed coordinate frame.
 - Verify whether changing the order of transformations changes the result.
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Expected Learning Outcome

After completing this experiment, students should be able to:

- Explain the significance of homogeneous transformation matrices.
 - Perform planar and spatial translations.
 - Perform rotations about X, Y, and Z axes.
 - Combine multiple transformations.
 - Understand the effect of transformation order.
 - Visualize robotic coordinate frames.
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